

Lab 05 – Greenfield Plant Architecture Workshop

Industrial Security (Master) • Maps to Section 05 • 4 hours, group of 4

LAB 05**🎯 Goal**

Produce a complete secure-by-design package for a fictional greenfield plant: zones & conduits, vendor SDL requirements, ATP, and a risk register.

✂ Step Scenario

A new 50-MW biogas plant. Single site, one DCS, one SIS, one HMI room, one Internet uplink for vendor remote support and a daily energy-trading data export to corporate IT.

✂ Step 1 – Zones & conduits diagram

On a single sheet or in draw.io, draw the architecture. Label every level (Purdue 0–5 + iDMZ at 3.5). Assign target SLs. For every conduit, write source / destination / protocol / port / direction. Mark the data-diode candidates.

✂ Step 2 – Vendor SDL clauses

Pick one component (PLC, HMI, historian). Write the 5–10 contractual clauses that the procurement team will paste into the RFQ.

✂ Step 3 – Acceptance Test Procedure

For the same component, write the on-arrival ATP. Each step: action, expected result, evidence.

✂ Step 4 – Risk register seed

List ten plausible risks for the plant; score with a simple Lik×Imp matrix; propose owners.

📁 Deliverable

A single PDF containing: the architecture diagram, conduit rules, SDL clauses, ATP, and the seed risk register.

⚠ Caution

Fictional plant only – do not reuse a real client's IP ranges or asset list, even partially.